TENTATIVE Syllabus v.1

Instructor: Dr. Roger Stanev, rogerst@uw.edu

Class: Mon/Wed 8:45-10:45 am at UW2 240 or via Zoom. Password: 382A

Join URL: [TBD]

Office Hours: [TBD] contact me via email and signup via Canvas Calendar

Class Discord: https://discord.gg/KxZ8wMdz

Course Description

Principal ideas and developments in artificial intelligence, such as problem solving, knowledge representation, search, reasoning under uncertainty, and learning.

Course Learning Goals

Outcomes:

1. an ability to apply knowledge of mathematics, science, and engineering;

2. an ability to design a system, component, or process to meet desired needs within

3. realistic constraints such as economic, environmental, social, political, ethical, health and safety, manufacture-ability, and sustainability;

4. an ability to use the techniques, skills, and modern engineering tools;

5. knowledge of contemporary issues;

6. a recognition of the need for, and an ability to engage in lifelong learning;

7. an ability to identify, formulate, and solve engineering problems; and

8. an ability to design and conduct experiments, as well as to analyze and interpret data

At the end of this course, students will be able to:

1. Describe what artificial intelligence (AI) means and how machines can process information intelligently;

2. Identify the different fields that comprise AI as well as techniques and heuristics for each area;

3. Create a formal representation of a complex problem; and

4. Write programs to solve complex problems using AI methods in areas such as search, reasoning, natural language processing, games, and robotics.

Textbooks

[Reference] Russell, Stuart J., and Norvig, Peter. *Artificial Intelligence : a Modern Approach*. 4th edition, Pearson, 2020. http://aima.cs.berkeley.edu/ (Links to an external site.) and https://www.amazon.com/Artificial-Intelligence-Approach-Stuart-Russell/dp/9332543518 (Links to an external site.)

[Recommended] Russell, Stuart J., *Human Compatible : Artificial Intelligence and the Problem of Control*. Penguin Books, 2019.

Grading

- Exercises: 15%

- Projects: 40%

- Midterm: 20%

- Final: 25%

A scale of 90s (3.5■4.0), 80s (2.5■3.4), 70s (1.5■2.4), 60s (0.5■1.4) is a rough guide. A student who achieves 75% of the possible points will receive a 2.0 grade in this course. Students with 95% and above will receive a 4.0 grade. Scores in between will be interpolated.

Distance Learning

Owing to the UW response to COVID-19, this course will be delivered partially online through the Canvas learning management system and Zoom (supplemented by other web resources). This means that you may need to put in more time on your own than in a conventional course. Of course, the benefit of this structure is that you do not need to be physically present at UW-Bothell.

This course is scheduled to run synchronously at your scheduled class time via Zoom.
Policies

Assignment Submission:

For projects, you can work individually or in pairs. When working as a pair, all substantial work must be performed with both students present (virtually). You can submit one and only one project 24 hours late without any penalty (if working as a pair both people will need to use their one-time extension). If you are using your 24 hour extension, post a brief explanation to "Using Extension" assignment.

Other than when working in pairs, talking about code is OK, copying each others code is typically not OK. Looking at references to understand how a functions gets used is OK; looking up assignment solutions is not OK. It is also not acceptable for your code or solutions to be publicly accessible on the web (for example, a public GitHub repository). Plagiarism will result in an assignment score of zero and a misconduct letter in your student record. Please be very careful to adhere to the student code of conduct: http://www.washington.edu/cssc/for-students/student-code-of-conduct/ I will make allowances for exceptional circumstances such as sickness, bereavement and official university business. I will not make exceptions for work, other classes, personal obligations, etc.

Attendance: Attend all classes. You are responsible for all the material covered in class, as well as any announcements including change of due dates or assignment specifications. There will also be graded in-class group exercises to practice problem solving. If you miss a class, I expect you to make-up for it on your own by asking your friends, reviewing the textbook, lecture materials, etc.

Communication: Use your UW email rather than "Canvas Messaging" to communicate directly with me. "Canvas Submission Comments" should only be used to draw the grader's attention to a specific part of your submission.

Problems: If you are having difficulties, write to me. If I don’t know about it, I cannot help you. Small problems can be fixed easily early in the quarter, but might become impossible to fix later on.

Course Material: Lecture notes and other material will be posted to Canvas under "Files".

See the School of STEM Course Policies, which covers:

- Academic Integrity
- Access and Accommodations
- Classroom Emergency Preparedness
- For Our Veterans
- Grade of Incomplete
- Inclement Weather
- Parenting Resources

- Religious Accommodations
- Respect for Diversity
- Student Support Services
- Surviving Sexual and Relationship Violence
- Wonder How to Address Faculty?

Course Calendar

Week	Tuesday/Thursday	Notes
09-11 Jan	Introduction, Uninformed Search	
16 Jan 18 Jan	A* Search and Heuristics	**Search Project**
23 Jan 25 Jan	Constraint Satisfaction	
30 Jan 01 Feb	Minimax, Expectimax, Utilities	**Multi-Agent Search Project A**
06-08 Feb	**Midterm**	
13 Feb 15 Feb	Markov Decision Processes	**Multi-Agent Search Project B**
22 Feb 27 Feb	Uncertain Knowledge & Reasoning	
01 Mar 06 Mar	Bayes' Nets, Ethics & Safey	**BN Project**

13-15 Mar	**Finals**	No classes during finals week